5G/ LTE/ 6G

7. Application	User interaction	HTTP, SMTP, FTP	**Data**
6. Presentation	Encoding, encryption	SSL, JPEG	**Data**
5. Session	Manages sessions	SIP, RPC	**Data**
4. Transport	Reliable/unreliable transmission	TCP, UDP	**Segment**
3. Network	Routing, IP addressing	IP, ICMP, RIP	**Packet**
2. Data Link	MAC addressing, framing	Ethernet, Wi-Fi	**Frame**
1. Physical	Signal transmission	Fiber, Copper, Radio	**Bits**

Streaming network telemetry is a real-time data collection service in which network devices, such as routers, switches and firewalls, continuously push data related to the network's health to a centralized location.

telemetry is the science of transmitting measurements from a remote source to a receiving station for storage and analysis

Data streams with protocols such as TCP, UDP, XML-RPC or gRPC in one of two ways
cadence-based telemetry, is when data sends information to a collector at regularly configured intervals.
event-based telemetry, is when data sends when a specific event occurs, like when a critical link goes down or when the throughput on a link passes a specified maximum threshold.

SNMP uses a pull technology where an SNMP collector had to request telemetry information at regular intervals.

SNMP vs. Telemetry

	SNMP	Telemetry
HOW IT WORKS	Polling mechanism collects device performance data and returns data to management platform	Push model continuously sends device operational data to management system
PROTOCOLS USED	User Datagram Protocol	User Datagram Protocol or TCP
USE CASES	Retrieving static data, such as inventory or neighboring devices	Collecting high-resolution performance data, such as high-speed network interface statistics
BENEFITS	Simple protocol and easy to perform ad hoc data collection; widely supported by network devices and monitoring platforms	Sends data at higher rate; more efficient and practical
CHALLENGES	Management system repeatedly creates and sends data requests to each device	Telemetry that relies on TCP connections can use large amounts of memory

Next, an administrator can selectively choose what information they want to collect. Standard data models such as IETF YANG, OpenConfig or vendor proprietary models help accomplish this process. Finally, admins choose the frequency that cadence-based data streams to a collector.

Communication networks rely on several fundamental concepts to ensure efficient data transmission between devices. Here are some key concepts:

1. **Packet Switching**
    - Data is broken into smaller units called packets.
    - Each packet contains a header (with source, destination, and control information) and a payload (the actual data).
    - Packets are transmitted independently and reassembled at the destination.
2. **Routing**
    - The process of selecting the best path for data packets to travel across a network.
    - Routing algorithms determine the optimal route based on metrics like hop count, latency, and bandwidth.
    - Common routing protocols: OSPF, RIP, BGP.
3. **Packet Forwarding**
    - The actual process of moving packets from one node to another based on routing table decisions.
    - Involves looking up the destination address and sending the packet to the next hop.
4. **Network Topology**

- The physical or logical arrangement of network devices.
- Examples: Bus, Ring, Star, Mesh, Hybrid.

5. **Addressing**
   - Uniquely identifying devices in a network using addresses.
   - Types: MAC addresses (Data Link Layer), IP addresses (Network Layer).
6. **Encapsulation & Decapsulation**
   - Encapsulation: Wrapping data with necessary protocol headers as it moves down the OSI model.
   - Decapsulation: Stripping headers as data moves up the OSI model.
7. **Flow Control**
   - Managing data transmission rate to prevent congestion.
   - Protocols like TCP use techniques such as sliding window.
8. **Congestion Control**
   - Preventing or managing excessive network traffic to avoid packet loss and delays.
   - TCP congestion control algorithms: Slow Start, AIMD.
9. **Quality of Service (QoS)**
   - Ensuring priority-based handling of different types of traffic.
   - Used in VoIP, video streaming, and real-time applications.
10. **Security & Encryption**

- Ensuring data confidentiality, integrity, and authenticity.
- Protocols: TLS, IPSec, WPA for wireless security.

Would you like a deeper explanation of any of these?

The **core network** in **4G (LTE)** and **5G (NR)** is the backbone of mobile communication, handling **authentication, data routing, mobility management, security, and internet access**. Let's break it down:

---

1. What is the Core Network?

- The core network is a **centralized system** that connects **radio networks (cell towers)** to the **internet, voice networks, and other mobile users**.
- It manages **data flow, user identity, mobility, security, and quality of service (QoS)**.
- In **4G LTE**, it is called the **Evolved Packet Core (EPC)**.
- In **5G**, it is called the **5G Core (5GC)** and is more advanced, flexible, and cloud-based.

---

2. Where is the Core Network Located?

- The core network is typically hosted in **data centers** operated by mobile network operators (e.g., AT&T, Verizon, Vodafone).
- Some parts of the core network may be **distributed** across multiple locations for redundancy and low latency.
- In **5G**, the core network can be **cloud-based (virtualized)**, running on edge computing systems **closer to users** for lower latency.

---

3. 4G LTE Core Network (EPC) Components

The **Evolved Packet Core (EPC)** in 4G consists of the following key elements:

(A) Mobility Management Entity (MME)

- Manages user connections, authentication, and mobility.
- Assigns users to the best cell tower when moving.
- Works with the **Home Subscriber Server (HSS)** to authenticate users.

(B) Home Subscriber Server (HSS)

- Central database storing user profiles, SIM details, and authentication info.
- Works with the **MME** to verify subscriber access.

(C) Serving Gateway (SGW)

- Routes user data between the base stations (eNodeBs) and the Packet Data Network Gateway (PGW).
- Handles user mobility between towers.

(D) Packet Data Network Gateway (PGW)

- Connects users to the **internet** and external networks.
- Manages IP address allocation, Quality of Service (QoS), and firewall/security.

(E) Policy and Charging Rules Function (PCRF)

- Controls user data speeds, billing, and network policies.
- Ensures network fairness based on subscription plans.

---

4. 5G Core Network (5GC)

The **5G Core (5GC)** is more advanced than the **EPC** because it is **fully cloud-native, software-defined, and service-based**. It has the following components:

(A) Access and Mobility Management Function (AMF)

- Equivalent to **MME** in 4G.
- Handles authentication and mobility between cells.

(B) User Plane Function (UPF)

- Replaces **SGW + PGW** from 4G.
- Routes internet data, ensuring low latency.
- Can be **deployed closer to the user (edge computing)** for real-time applications.

(C) Session Management Function (SMF)

- Manages IP address assignment and user sessions.
- Handles Quality of Service (QoS) and prioritization of traffic (e.g., emergency calls, gaming, video streaming).

(D) Network Repository Function (NRF)

- A new function in **5G**, allowing **different network components to discover and communicate with each other** dynamically.

(E) Unified Data Management (UDM)

- Equivalent to **HSS** in 4G.
- Manages user authentication and profiles.

(F) Network Slice Selection Function (NSSF)

- Allows **"network slicing"**, where different virtual networks serve different purposes (e.g., one for gaming, one for IoT, one for healthcare, etc.).

---

5. Key Differences Between 4G EPC and 5G Core

Feature	**4G EPC**	**5G Core (5GC)**
Architecture	Centralized	Distributed & Cloud-based
Latency	~20-50 ms	<10 ms
Virtualization	Partial	Fully Virtualized
Network Slicing	No	Yes
Internet Handling	Centralized via PGW	Edge-based via UPF
IoT Support	Limited	Massive IoT Support

---

6. Where Does the Core Network Run?

- **4G EPC** is typically run in **centralized data centers**.

- **5G Core** is deployed in **cloud infrastructure**, meaning:
  - Some functions run in **large data centers** (like AWS, Google Cloud, or private telco clouds).
  - Some functions run on **edge servers** close to the users (e.g., inside cities or stadiums for low-latency services).

---

7. Summary

- **4G EPC** is **centralized**, designed for high-speed mobile broadband.
- **5G Core (5GC)** is **cloud-native, virtualized, and supports ultra-low latency applications**.
- 5G introduces **network slicing**, allowing different services (IoT, streaming, emergency services) to have their own dedicated resources.

Would you like details on **how calls or data flow through the core network**? 🚀

- Channel characteristics - OFDM - ICI
- Fading channel
- OTFS
- coherent and non-coherent transeviers
- adaptive OFDM
- GPS based MIMO -OFDm perform - as fn of distance from base station
- Pareto Optimal Networking
- Energy efficiency → cost function evaluation
- Polar / coding schemes - LDPC - 5G border control channel
- Path/link lifetime

Data Flow in a Packet-Switched Satellite Network: Hops, Forwarding, Routing, and Switching

In a **packet-switched satellite network**, data is transmitted as small, independent units called **packets**. Each packet contains:

- A **header** (source, destination, sequence number, error checking)
- A **payload** (actual data)
- A **trailer** (optional error-checking)

To understand how packets move through the network, we need to explore **packet hops, forwarding, routing, and switching**, especially in **satellite-based systems like Iridium and Starlink**.

---

1. Packet Hops in a Satellite Network

A **hop** refers to the movement of a data packet from one network node to another. In a satellite network, packet hops can occur in several ways:

1. **User → Satellite (Uplink Hop)**
   - A packet is sent from a ground terminal (e.g., a ship, aircraft, or phone) to the nearest satellite.
2. **Satellite → Satellite (Inter-Satellite Hop using ISLs)**
   - The satellite processes the packet and, if the destination is far away, forwards it to another satellite via **Inter-Satellite Links (ISLs)**.
   - This step can involve multiple hops between satellites, forming a **space-based mesh network**.
3. **Satellite → Ground Station (Downlink Hop)**
   - Once the packet reaches a satellite close to the destination, it is sent down to a ground station connected to the internet or a private network.
4. **Ground Station → Final Device (Terrestrial Hop)**
   - The packet moves from the ground station through fiber optics, cellular, or Wi-Fi networks to reach the final recipient.

💡 **Example**:

A message sent from a ship in the Pacific to a command center in New York may go through:

- **Uplink hop** (Ship → Iridium satellite)
- **ISL hops** (Satellite-to-satellite transfers in space)
- **Downlink hop** (Satellite → U.S. ground station)
- **Terrestrial hop** (Ground station → New York office)

---

2. Packet Forwarding in a Satellite Network

Packet **forwarding** refers to the process where a node (satellite or router) receives a packet and decides where to send it next.

🔷 **Key Packet Forwarding Techniques in Satellites**:

1. **Hop-by-Hop Forwarding (Most Common in LEO Networks)**
   - Each satellite **processes** the packet and forwards it to the next best satellite or ground station.
   - **Used in Iridium's ISL routing**: packets hop from satellite to satellite.
2. **Source-Based Forwarding (Less Common)**
   - The packet's **entire route is determined at the source** before transmission.
   - Used when satellites have **limited processing capability** (e.g., GEO satellites).
3. **Cut-Through Forwarding (High-Speed)**

- Instead of waiting for the full packet, satellites start forwarding as soon as they receive part of it.
- **Reduces delay** but requires high-speed links and error handling.

💡 **Example in Iridium**:

1. A ground terminal sends a packet to an Iridium satellite.
2. The satellite **analyzes the destination address** and forwards it to a neighboring satellite using ISLs.
3. The process repeats until the packet reaches the best satellite for downlink.

---

3. Packet Routing in a Satellite Network

Packet routing is the process of determining the **best path** a packet should take from source to destination.

Types of Routing Used in Satellite Networks

🔷 1. Static Routing (Common in GEO Networks)

- Routes are **predefined** and do not change dynamically.
- Used in **GEO satellite systems** where each satellite has a **fixed footprint**.

🔷 2. Dynamic Routing (Used in LEO Networks like Iridium & Starlink)

- **Satellites calculate the best route in real-time** based on congestion, link quality, and satellite positions.
- Uses **dynamic algorithms like OSPF (Open Shortest Path First) or BGP (Border Gateway Protocol)**.

🔷 3. Predictive Routing (Unique to LEO Satellites)

- **Accounts for satellite movement** and predicts the best paths **before satellites move out of range**.
- Iridium and Starlink use this to ensure seamless handoffs.

💡 **Example of Routing in Iridium**:

1. A satellite receives a packet from a user terminal.
2. The satellite **checks its routing table** (stored in onboard memory).
3. If the destination is **in another region**, it forwards the packet via ISLs.
4. If the destination is nearby, it sends the packet directly to a ground station.

---

4. Packet Switching in a Satellite Network

Packet Switching is the method of transmitting packets independently, allowing them to take different paths and be reassembled at the destination.

🔷 **Two Main Packet-Switching Techniques in Satellites**

1. Datagram Packet Switching (Used in Iridium & Starlink)

- Each packet is **routed independently** and may take different paths.
- The destination **reassembles packets in the correct order**.
- **Pros**: More fault-tolerant, adapts to network congestion.
- **Cons**: Requires more processing at the receiver.

2. Virtual Circuit Packet Switching (Used in Some GEO Networks)

- A **predefined path (virtual circuit)** is established before sending packets.
- **Pros**: Ensures packets arrive in order.
- **Cons**: Less flexible, not ideal for dynamic satellite networks.

💡 **Example of Packet Switching in Starlink**:

1. A user in Alaska sends a request to a website.
2. The request is broken into **packets** and sent to the nearest satellite.
3. **Some packets travel directly to the ground station**, while others hop between satellites.
4. The website's response follows the **most efficient path available** at that moment.
5. Packets are **reassembled at the user's terminal**.

---

5. End-to-End Example: How a Packet Travels in a LEO Satellite Network

🔷 **Scenario**: Sending an email from a ship in the Pacific to a laptop in London using Iridium.

1️⃣ **Ship sends an email** → The email is broken into **packets** and sent to the nearest **Iridium satellite** (Uplink).

2️⃣ **Satellite checks routing table** → It decides whether to:

- Send packets **to another satellite using ISLs**, or
- Forward packets **to a ground station**.

3️⃣ **Packets hop through satellites** → They follow the **shortest and least congested path** via ISLs.

4️⃣ **Packets reach the best ground station** → A satellite over the UK forwards the email to an Iridium **ground station in Europe**.

5️⃣ **Packets enter the terrestrial network** → The ground station sends them over the **internet** to London.

6️⃣ **Laptop receives and reassembles packets** → The email appears complete in the inbox.

---

6. Conclusion: Why OBP and ISLs Matter for Efficient Packet Handling

Packet switching, forwarding, routing, and hops are **optimized by Onboard Processing (OBP) and Inter-Satellite Links (ISLs)** in LEO satellite networks like Iridium and Starlink. This results in:

✅

Faster data transmission (less reliance on ground stations).

✅ **Lower latency** (packets travel through space instead of long ground routes).

✅ **Global connectivity** (especially over oceans and polar regions).

✅ **Improved reliability** (multiple paths exist if one fails).

Would you like a comparison of **Iridium vs. Starlink's ISL packet routing**, or a latency breakdown? 🚀

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School Project

SKYWAN 5G Feature Overview

New Modem Features:

- Always dual demodulator (TDMA + DVB-S2)

- 16APSK for TDMA
- ACM for DVB-S2
- Extended L-Band range (for Ka-Band and to cover all 3 Ku-band ranges)
- LNB control supported by IDU (DiSEqC)

Improved IP Features

- Four independent 1Gbit/s Ethernet ports
- VRF (Virtual Routing & Forwarding)
- VLAN, DNS-Proxy, DHCP, NAT
- IPv4, IPv6
- OSPF, BGP, GRE

Modular Terminals

- Scalable with network traffic growth
- Cascading at Hub and Remote Sites

Network Topology Flexibility

- Lowest price for small networks
- Multi-hub / Sub Network Hubs
- Same Hardware everywhere
- Sub networks sharing the same capacity
- Single hop wherever required
- Support of multiple DVB-S2 outbound locations

User Data Rates

- TDMA from 200 K symbols/s up to12 M symbols/s, i.e. up to 20 Mbps rates in bidirectional usage (20/20 Mbps)
- Total data rates per single remote: 75/20 Mbps
- Total data rates per cascaded remote: 75/80 Mbps
- multiple DVB-S2 outbound paths from different locations per network

Multi-Master / Multi-Hub

- Each terminal can fulfil “TDMA master” functionality
- Seamless failover TDMA redundancy concept
- Hub functionality by licence key

Network Management

- Simple installation & management

- transaction based reconfigurations without terminal reboots
- secure management interfaces only

CISCO-style CLI with transaction mechanisms

Regulations

Yes, Wi-Fi modules are designed and configured differently for various countries due to differences in regulatory standards, frequency allocations, and maximum allowable transmission power. These variations ensure compliance with local laws and minimize interference with other wireless systems.

Key Differences in Wi-Fi Modules Across Countries

1. **Frequency Bands and Channels**:
   - Different countries allocate specific frequency bands and channels for Wi-Fi usage. For example, the 2.4 GHz and 5 GHz bands are commonly used worldwide, but the number of available channels and their usage rules vary by region. Some countries also allow the newer 6 GHz band (Wi-Fi 6E), while others do not[1][4][5].
2. **Transmission Power Limits**:
   - Maximum transmission power (measured in EIRP) is regulated to prevent interference. For instance:
     - The U.S. allows higher power levels (e.g., up to 1000 mW for certain channels in the 2.4 GHz band).
     - European countries typically have stricter limits (e.g., 100 mW for the same band)[1][5].
   - Devices must comply with these limits, which may affect range and performance.
3. **Dynamic Frequency Selection (DFS) and Transmit Power Control (TPC)**:
   - In some regions, devices operating in specific parts of the 5 GHz band must implement DFS to avoid interfering with radar systems and TPC to manage power levels dynamically. For example:
     - DFS and TPC are mandatory in Europe for certain 5 GHz channels.
     - In Brazil, similar requirements exist but with slight variations[5].
4. **Country-Specific Certifications**:
   - Wi-Fi devices must pass local certification processes to be sold or used legally. Common certifications include:
     - FCC (United States)
     - CE (European Union)
     - ARIB (Japan)
   - Some countries require additional testing or documentation beyond these widely accepted certifications[2][3][4].

5. **Region-Locked Firmware**:
   - Many Wi-Fi devices come with firmware that restricts their operation based on the region selected during setup. This ensures compliance with local regulations but may limit functionality if the device is moved to another country[1][3].

Practical Implications

- Manufacturers often design Wi-Fi modules to support multiple configurations so they can be adapted for different markets through firmware or software updates.
- Users attempting to bypass regional restrictions (e.g., by selecting a different country in settings) may face legal issues or cause interference with other devices.

In summary, Wi-Fi modules are tailored to meet the specific regulatory requirements of each country or region, ensuring safe and efficient operation within local constraints.

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